Variables, Types and Functions

Python is a popular programming language for data science. A variable is a name that refers to a value, for example age = 25. Common built-in types are int, float, str, bool, list, tuple, set and dict.

A function is defined with the def keyword. Functions group reusable logic and can return values. Example: def square(x): return x * x.

Lists, Dictionaries and Comprehensions

A list is an ordered, changeable collection written with square brackets. A dictionary stores key-value pairs written with curly braces. A list comprehension builds a new list in one line, for example squares = [x * x for x in range(5)].

Exceptions are handled with try and except blocks. Files are best opened with the with statement, which closes the file automatically even if an error occurs.